

Lab Report: Deploying Gitea with Docker and Verifying Git LFS

Student Name: Yuanzhi Tang

1. Objective

To learn how to deploy a private Git repository using Gitea via Docker.

To verify that Gitea supports Git LFS (Large File Storage).

To upload a 1GB large file through Git LFS and confirm successful storage.

2. Environment and Tools

Operating System: Windows 11

Docker Desktop: latest version

Gitea: latest Docker image

Git: 2.54.0

Git LFS: included with Git

3. Procedures

3.1 Prepare Docker and Deploy Gitea

Enable WSL2 and update the system environment.

Start Docker Desktop and confirm the service is running.

Run the Gitea container using the command:

```
docker run -d --name=gitea -p 3000:3000 -p 222:22 -v gitea_data:/data gitea/gitea:latest
```

Verify the container status with docker ps.

Access <http://localhost:3000> in a browser, complete the setup wizard, and create an administrator account named yixian.

3.2 Create a Repository and Check LFS Support

Log in to Gitea and create a new repository named cloud-computing.

Go to the repository Settings page and open the LFS tab. Confirm that Git LFS is enabled by default.

3.3 Configure Git and Git LFS Locally

Install Git on the local machine.

Open Git Bash and verify the installation with:

```
git --version
```

```
git lfs install
```

Generate a 1GB test file using:

```
fsutil file createnew bigfile.bin 1073741824
```

3.4 Commit and Push the Large File via LFS

Initialize a new Git repository in the working directory.

Configure Git LFS to track the .bin file extension.

Add the .gitattributes file and the 1GB test file to the repository.

Set the Git user identity using:

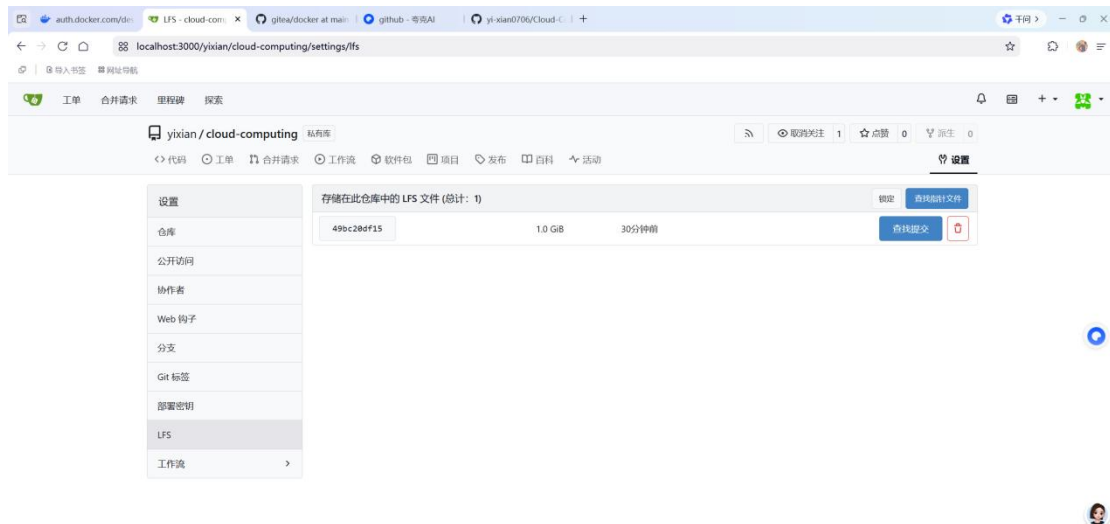
```
git config --global user.name "yixian"
```

```
git config --global user.email "2631024931@qq.com"
```

Commit the files with a descriptive message.

Link the local repository to the remote Gitea repository and push the changes. The output shows that the 1GB LFS object is uploaded successfully.

4. Results



```
由 Gitea 强力驱动 版本: 1.25.5 页面: 14ms 模板: 7ms
简体中文 许可证 API

Lenovo@DESKTOP-FQKLVLB MINGW64 ~
$ git --version
git version 2.54.0.windows.1

Lenovo@DESKTOP-FQKLVLB MINGW64 ~
$ git lfs install
Git LFS initialized.

Lenovo@DESKTOP-FQKLVLB MINGW64 ~
$ mkdir gitea-lfs-test && cd gitea-lfs-test

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test
$ fsutil file createnew bigfile.bin 1073741824
已创建文件 C:\Users\Lenovo\gitea-lfs-test\bigfile.bin

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test
$ git init
Initialized empty Git repository in C:/Users/Lenovo/gitea-lfs-test/.git/

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git lfs track "*.bin"
Tracking "*.bin"

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git add .gitattributes

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git add bigfile.bin

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git commit -m "Add 1GB test file for LFS verification"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Lenovo@DESKTOP-FQKLVLB.(none)')
```

```

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git config --global user.name "yixian"

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git config --global user.email "2631024931@qq.com"

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git commit -m "Add 1GB test file for LFS verification"
[master (root-commit) c34af8e] Add 1GB test file for LFS verification
 2 files changed, 4 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 bigfile.bin

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git commit -m "Add 1GB test file for LFS verification"
On branch master
nothing to commit, working tree clean

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git remote add origin http://localhost:3000/yixian/cloud-computing.git

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)
$ git push -u origin master
warning: use of unencrypted HTTP remote URLs is not recommended; see https://aka.ms/gcm/unsaf
feremotes for more information.
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/yixian/cloud-computing.git/info/lfs.locksverify tru
e
Uploading LFS objects: 100% (1/1), 1.1 GB | 150 MB/s, done.
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 431 bytes | 431.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/yixian/cloud-computing.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Lenovo@DESKTOP-FQKLVLB MINGW64 ~/gitea-lfs-test (master)

```


[工单](#)
[合并请求](#)
[里程碑](#)
[探索](#)


[yixian / cloud-computing](#)
私有库
取消关注 1
点赞 0
派生 0

[<> 代码](#)
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1 提交
1 分支
0 Git 标签

搜索代码...

master
转到文件
添加文件
<> 代码

	yixian c34af8e12b	Add 1GB test file for LFS verification	1小时前
	.gitattributes	Add 1GB test file for LFS verification	1小时前
	bigfile.bin	Add 1GB test file for LFS verification	1小时前

描述

无详细信息

管理主题

1 GiB

由 Gitea 强力驱动 版本: 1.25.5 页面: 57ms 模板: 7ms

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